

PHARMACOLOGY

NCLEX STUDY GUIDE



Organized for study and printing | 53 content pages

Pharmacology

G1 Respiratory Medication

❑ **GRUPO#1 . Bronchodilators.** - Used for asthma, COPD. Abren el pulmon pero aprietan el corazon.

1. β adrenergics Agonists
2. Xanthine derivates
3. Anticholinergics

1. β Adrenergic Agonist

1. SABA Short acting β agonist	2. LABA Long acting β agonist
- Use for acute asthma attack ❑ Albuterol ❑ Terbutaline (use for preterm labor=use as a tocolytic) para para el parto y no paran antes de tiempo) . No dar en embarazada con diabetic bc increase blood sugar) No in diabetic bc increase blood sugar	- Use for chronic Asthma ❑ Bambuterol ❑ Salmeterol ❑ formoterol Use to prevent crisis No in diabetic bc increase blood sugar
3. Epinephrine - Use for Anaphylaxis - In emergency - Asystole Epinephrine others Side effect Skin necrosis (bc dilata el pulmón pero aprieta el corazón) - o Antidote de la skin necrosis = Phentolamine = SQ (↓ BP mismo tto for pheochromocytoma)	Nota: #1 choice for Asthma = Albuterol

Side effect (de todos ellos tanto saba como laba) todo increase)

1. Insomnia
2. Restlessness
3. Cardiac stimulation (tachycardia)
4. hyperglycemia (no in diabetes)

5. Tremors
6. Headache
7. Anorexia, nausea, vomiting

NOTA:

EROL= Asthma (suben el blood glucose)

LOL= Blood pressure (mask hypoglycemic sides) ninguno de ellos en diabetic

2. Xanthine Derivates

- **Theophylline or aminophylline** = use for acute or chronic Asthma

Side effect

1. Nausea, vomiting, diarrhea
2. GERD, se contraindica aqui!
3. Tachycardia
4. Cardiac arrhythmias.
5. PVCs, supraventricular tachycardia
6. Tremors, nervousness, restlessness
7. Increase BP, increase Glucose.

Nota: los side effects the bronchodilator es para arriba.

Therapeutic Level

- 10-20 mcg/ml
- More than 20 = Stop medication and call the Dr.= toxicity
 - o Signs of toxicity
 - ☑ Tachycardia
 - ☑ Restlessness
 - ☑ Neurologic Symptoms

Contraindication

1. Hypertension and arrhythmias.
2. Seizure Disorders
3. GERD or peptic ulcers
4. No in Hyperthyroidism
5. No with allopurinol
6. No with Cimetidine (esta no se da con nadie)
7. No with Erythromycin. (use for allergic to penicillin)
8. No with influenza vaccine.
9. No with stimulants.

10. No with St John wort ↑ BP. (yerbita for depression = esperar dos semanas)
11. No in glaucoma
12. No with digoxin
13. No with phenytoin
14. No with MAOIS (bc increase Bp)
15. No with otc med (los del catarro)
16. No with gimsen (bc ↑ BP)
17. No with ephedra (bc increase bp) (gotas nasales para la presión)

Grupo3. Anticholinergics

Choice for COPD

Bronchodilator, tachycardia and mydriasis)

1. Tiotropium
2. Ipratropium

Side effects

- Dry Mouth
- Decrease Tear
- Decrease Salivation
- Decrease Sweating
- Decrease Urinary output
- Constipation
- No in glaucoma
- No in BPH
- No in peanut allergies patient

Grupo2. Glucocorticoids

Choice for Atsma

Inhaled (Spray por la boca)

1. Beclomethasone
2. Budesonide
3. Fluticasone
4. Trinacinolone

Parenteral

1. Prednisone
2. Prednisolone
3. hydrocortisone

- First administer a bronchodilator, and 5 min after administering a steroid
- Immediately after steroid use wash mouth to prevent Oral Trush (candidiasis) y bota el enjuague
- Si ya tienes candidiasis

Tx: Nystatin mouth wash, **rinse and swallow the mouth was**

Nota Fluticasone es diferente de fluconazole, fluconazole es para los hongos

Side Effect (for both)

- Like Cushing Syndrome = too much steroid
- ↑ Blood glucose and Bp
- Osteoporosis
- Healing retardation
- ↑ weight, fluid retention
- Immunosuppression
- Cataracts, glaucoma
- In children – growth retardation
- **Tapering slowly to prevent Addisonian crisis (quitarlo suave)**
- Take in the morning with food.

NOTA: (never after 6:00 pm to prevent insomnia.

Grupo 3. Leukotriene Modifier

1. Montelukast -Use for chronic asthma = Preventive.
2. Zafirlukast -Take 1h before 2 h after meals

Side effect

1. Headache
2. **Liver toxicity** (No in liver disease = check for liver function during treatment).
3. No in breast feeding woman
4. Flu like symptoms

Grupo 4. Monoclonal Antibodies (mab)

Omalizumab = Chronic Asthma prevention

Palivizumab = Bronchitis Prevention

Note: **Infliximab=board** - use for Rheumatoid Arthritis

Adalimumab: arthritis

Ulcerative colitis

Crohn disease

Only use for prevention.

- No in acute Tb

- No with Live virus Vaccine
- No with immunoglobulins.
- No in infections actives

Side Effect

- First dose administers under supervision to check for **Anaphylactic reaction** =sq injections
- Risk for Cancer

Grupo 5. Nasal Decongestant

- Use for rhinitis and common cold, allergies.

- Glucocorticoids Budesonide Fluticasone Triamcinolone Beclomethasone SAME S/E OF Glucocorticoids <u>Side Effect. For both.</u> - Rebound rhinitis = Use only for 72 hours to prevent rhinitis. - Only effective for 72h no usarlo por mas de 72h	- Non-Glucocorticoids 1. Naphazoline 2. Phenyl epinephrine 3. Pseudo epinephrine <u>Side Effect. For both.</u> - Rebound rhinitis = Use only for 72 hours to prevent rhinitis. Side effect for Non-gluco - ↑ BP - Tachycardia - Insomnia - Nervousness - ↑ Blood glucose - No with ma juan Ephedra (bc es lo mismo) - - No with MAOIs - No with Ginseng - No with Stimulants = caffeine, cocaine
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Grupo 6. Expectorants and mucolytic Agents

1-Guaifenesin (Mucinex)	2-Acetylcysteine (Mucolytic)
- Use for dry cough *Take with plenty fluid	- Use to make cough more productive - Use after bronchodilators, 5 min after acetylcysteine, and 30 min after chest physiotherapy= tx most important to sacar todos los mocos trabados de la cystic fibrosis. = TAKE with plenty fluid.

	- Choice for Cystic Fibrosis - Rotten eggs Odors = Expected (huele a podrido) - Use as Antidote in Tylenol intoxication. Nota: Tylenol more than 4g/day = toxicity

Grupo 7. Antituberculosis medication

Board" Complete all treatment = 6-9 months even without symptoms

First Line	Second line = se pueden mezclar
- Isoniazid - Rifampin - Ethambutol - Pyrazinamide Side Effect -Hepatotoxicity= if *Jaundice *Clay feces *Dark urine *Pruritus Stop. Medication + call the Dr.	- Streptomycin - Amikacin - Aminoglycosite - Levofloxacin or - Ciprofloxacin Side effect of aminoglycosides - Ototoxicity - Renal Toxicity - Elevado el bun y la creatinine or tengo tinnitus = stop and call dr) (dañan todo except liver) Side effect of levofloxacin & ciprofloxacin - Achilles Tendon rupture = If Heel pain = Stop medication + call Dr. NOTE: statins son los que dan calf pain

Medication	Side effect & Considerations
1. Isoniazid	- Peripheral neuropathy - Give with Vit B6 to prevent toxicity

2. Rifampin	- Orange discoloration of body secretion = Expected do not call the Dr. Teaching: - Avoid use contact lenses - Decreased effect of Oral Contraceptive = Use two different contraceptive methods
3. Ethambutol	- ECTS - neuritis = te deja ciego - baseline visual acute before use - children less than 12 years - cataracts, glaucoma, or any visual - s - s - ination between red and green - (Daltonics)
4. Pyrazinamide	- ↑ Uric acid - -No in gout - No in Renal impairment - No in children - No with gout medication = - *Allopurinol * Colchicine *Probenecid - Thrombocytopenia = decrease platelets (Risk for bleeding)

Grupo 8. Antitussive medication (cough suppressant)

Opioids	Nonopioids
- Codeine phosphate - Codeine sulfate - Hydrocodone	Dextromethorphan
<u>Side effect</u> 1. Drowsiness, sedation, dizziness 2. GI irritation, nausea 3. Dry mouth 4. BOARD: Constipation – Antidote= Sennoside 5. BOARD: Respiratory depression – Antidote = Naloxone Naltrexone = antidote for Opioids + alcohol (T – Tomar) <u>Intervention</u> - Plenty fluid - Risk for fall (evitar en viejitos)	

Grupo 9. Antihistaminic

1st Generation (+ sedation)	2nd Generation (Less sedation, no somnolence)
1. Diphenhydramine=Benadryl 2. Chlorpheniramine 3. Cyproheptadine= used for serotonin syndrome. 4. Dimenhydrinate = Grabinol Use for nausea and vomiting Use for Motion sickness = choice (Choice para-Mareo de Barcos)	1. Loratadine 2. Cetirizine

Nota apart**Scopolamine:** Antichorro=Anticholinergic**Patch:** behind the ear for motion sickness.

(apply 4h before subas al Barco, change every 12h)

Oral : (pastillas) o(goticas) use for googling sound
(Rattle sound) =dying Pt)**Side Effect =todos**

1. Somnolence
2. Drowsiness
3. Fatigue
4. Central Nervous System depression- (Avoid operating heavy machinery)
5. **Don't drive**
6. Anticholinergic side effect

No in glaucoma

No in BPH

No in Sjogren = bc dry mouth, dry eyes, dry vagina (decrease todo)

7. Administer with food and **milk.**
8. Photosensitivity = **use sun cream**
9. Mouth care = **Hard candies**
10. Avoid alcohol, sedatives
11. Don't drive

G2-Cardiovascular medication**Grupo 1. Anticoagulats medication**

(To prevent clots and MI)

Oral	Parenteral
- Warfarin. = PT-INR Contraindicated in pregnancy.	- Heparin = at Hospital. -APPT One week before discharge give two warfarin and heparin, and is ready to remove heparin = PT-INR 2-3 = Board
o Teratogenic = never use in pregnant	- Enoxaparin – At home = choice for pregnant, low molecular weight heparin - Fondaparinux - Dabigatran - Argatroban
Uses For: -To prevent clot formation, if clot is formed already don't work, and give thrombolytic therapy clots are present 1. MI 2. Unstable angina 3. Atrial fibrillation 4. DVT 5. Pulmonary embolism 6. Mechanical valve replacement	
Contraindicated In Active bleeding except Disseminated intravascular coagulation (DIC) = only bleeding treat with heparin	

Heparin Administration - Administer subcutaneous = $\angle 90^\circ$ hasta lo Amarillo.	NOTA: La insulin gives at 45 and if obese give at 90.
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Needle gauge = 25-27 Needle length = 0,5 – 0,6 – 5/8 If malnourish = $\leq 45^\circ$ <u>Side Effect</u> 1. Heparin induce thrombocytopenia (if platelets $\downarrow$ = Stop + Call Dr.) 2. Hematuria 3. Epistaxis 4. Ecchymosis 5. Bleeding gums 6. Hypotension <u>Control Lab (aPPT for normal person 25-35)</u> - Aptt = 45 – 75 with heparin = (document) - more than 75 stop medication + call Dr. bc risk for bleeding. - Less than 45 = Call Dr. to increase it. Note: Antidote of heparin = Protamine Sulfate	
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Warfarin

- Use for long term anticoagulation

Side effect

Same to heparin, Except for thrombocytopenia effect (just heparin).

Control labs. (Normal person without warfarin = 9-12)

- PT with warfarin = if more than 18 stop warfarin + Call Dr.

With warfarin is normal until 18. If more than 18= stop and call Dr.

- INR – 1.5-2 without warfarin normal person.

With Warfarin 2 – 3 = Normal, more than 3 = stop + Call Dr.

Less than 2 – Call Dr. to increase it (bc risk for MI).

- **NOTA: In case high dose of warfarin INR = 4, 5 = Normal and if**

1. Mechanical valve replacement. 4.5 normal too.

Diet

- Maintain constant or avoid green leafy vegetable.

-

Board: if both in the answer = Choose Maintain. Never increase.

One week before discharge administer:

-Heparin + Warfarin

- How do I know is ready to go home with warfarin = INR Normal = 2 -3.

-

Interaction (warfarin)

- | | |
|-------------------|------------------------------------|
| -No With (herb G) | - No With: |
| - Gingko | - Amiodarone |
| - Garlic | - Cimetidine. (no se da con Nadie) |
| - Ginseng | - Fluconazole |
| - Ginger | - Metronidazole |
| - St. John Wort | - Sulfa |
| - Fever few | |

Grupo 2de cardio. Thrombolytic Medication (to lysis of thrombus) = (PLASE)

- Alteplase
- Urokinase
- Recombinant Streptokinase
- Tenecteplase

Use for

- DVT = 5-7 days
- MI = 6 hours
- Ischemic stroke = 4.5 hours (before CT

Scan to confirm)

Recombinant Streptokinase

- No in history of β hemolytic Infraction
- Rheumatic Fever
- Endocarditis
- Glomerulonephritis.

Contraindication

1. Uncontrolled HTN

2. Prolonged or recently CPR
3. History of hepatic and renal disease
4. Active Internal bleeding
5. History of hemorrhagic stroke
6. Head trauma, ↑ICP
7. Recent surgery 2 months for:
 - Intracranial
 - Intraspinial
8. Recent surgery =10days for
 - Thoracic
 - Pelvic
 - Abdomen

Side effect. (thrombolytic)

1. Hemorrhage O Hemorrhage stroke

- ☐ Confusion
 - ☐ LOC Alteration
 - ☐ Slurred Speech
 - ☐ Paralysis
2. Cardiac dysrhythmias
 3. PVCs

Antidote Thrombolytic therapy = **Aminocaproic Acid** = Only use in life threatening: slurred speech or Hemorrhage.

Due effective if a PVC after the thrombolytic therapy and Call Dr

Nota: El tratamiento para un simple PVC = Quitar stimulants Call

Grupo 3 de cardio Antiplatelets

1. Aspirin
2. Clopidogrel
3. Dipyridamole
4. Ticlopidine
 - **Can administer at the same time with anticoagulant**

Side Effect

1. GI bleeding
 - Melena (tarry stools)
 - Hematemesis
2. Bruising
3. Hematuria

Aspirin-toxicity

1. Tinnitus = (sordera)
2. Metabolic acidosis = (Kussmaul respiration)
3. Bleeding

Treatment of aspirin toxicity.

Gastric lavage with charcoal activated

Nota: es el tto del que se empastilla PO

Grupo 4 de Cardio- Antihypertensive**Grupo 1- Diuretics**

1. Osmotic Diuretic
2. Carbonic Anhydrase Inhibitors
3. Loop Diuretics. **(USE IN EMERGENCY)**
4. Thiazide Diuretics. **(CHOICE FOR BP)**
5. Potassium Sparing Diuretic

1. Osmotic Diuretic

- Mannitol – use for brain edema to ↓ ICP

And can be used in the oliguric phase of acute kidney injury.

(Contraindicate when lung liquid)

Side Effect

1. Dehydration. **Not in sickle cell anemia bc is dehydration.**

2. ↓K – NO WITH DIGOXIN.

In low K don't give Furosemide

↓ Na = NO WITH LITHIUM

3. Administered through filter (bc crystals) and warm solution before administered. Don't give cold. Warm with a warmer pat.
4. S/E= CHF= lung crackles, SOB

Stop and Call dr bc manitol te saco el liquido de la cabeza y te lo puso para el pulmon, y ese S/E se tiene que reportar.

5. Seizure (contraindicate in epilepsy) no in seizures.

2. Carbonic Anhydrase Inhibitors

- Acetazolamide
- Dorzolamide
- Choice for glaucoma

Side Effect

1. Dehydration = ↓ Na, ↓ K (no with digoxin or with Lithium)
2. No in sulfa allergies patient. Also (Furosemide-Dorzolamide-Hydrochlorothiazide) son sulfa =no with sulfa allergies
3. No in liver or kidney dysfunction

3. Loop Diuretics

- Bumetanide
- Furosemide
- Torsemide
- Ethacrynic Acid

Use in acute edema.

CHF

Hypercalcemia

Renal insufficiency

Oliguric phase of acute kidney injury

Side Effect

(Loop diuretic) **Decrease all except Uric acid OJO**

1. Dehydration
2. ↓ Na, ↓ K, ↓ Ca, ↓ po4 (use for hypercalcemia)
3. ↑ Uric Acid – No in Gout
4. Da Thrombocytopenia = Risk of bleeding.
5. Orthostatic Hypotension
6. Photosensitivity
7. Thiamine deficiency (vit B1) = Wernicke (alcoholism)
8. Tinnitus, deafness (par cranial 8) when administer IV too fast – furosemide

Nota: Administer furosemide slow

- Never never Diuretics in sickle cell anemia bc dehydration, treatment for sickle cell = IV fluids
- Casi todos los medicamentos que dan orthostatic hypotension se dan en la noche, except por los loops diuretics = Furosemide bc orinan mucho en la noche, y se puede caer. Se administra en la manana.

4. Thiazide Diuretics (Choice for HTN (BOARD) never use in acute setting) (suben todo except K and Na)

- Chlorothiazide
- Hydrochlorothiazide
- Chlortalidone

Nota: Choice for home in HTN
Cross Allergies with Sulfa

Side Effect

1. ↓ K (no with Digoxin)
↓ Na (no with Lithium)
2. ↑ Uric Acid (No in gout)
3. ↑ Blood Glucose (No in diabetic) BOARD
4. ↑ Calcium (no in osteoporosis, no in kidney stone que sean calcium oxalate – take with plenty fluids)
5. Headache
6. Orthostatic Hypotension (= change position slowly)
7. Blood dyscrasia o ↓ WBC o ↓ RBC o ↓ PLATELETS
8. Photosensitivity (sun blocker)
9. Constipation

5. Potassium Sparing Diuretic

EKG Chance: Peak T Wave

- Spironolactone
- Amiloride

- Triamterene

Choice for ascites of liver cirrhosis

Use for HTN

Use for nephrotic syndrome

Contraindication

1. Kidney Failure
2. No with prils/Sartan (bc $\uparrow$ K) (decrease sodio no with lithium)

Side Effect

1. Increase K = Hyperkalemia (=tall T or peaked T wave)
2. Nausea, vomiting and diarrhea
3. Rash
4. Dizziness, weakness,
5. Headache
6. Anemia
7. Photosensitivity
8. $\downarrow$ Platelets

Just for spironolactone = (cambios hormonales)

1. Gynecomastia (Cenos en hombres) + galactorrhea (lechita por el peson)
2. Irregular menstruation
3. Amenorrhea

Intervention

1. Check for K during Treatment
2. Diet low in K

Nota: Cuales son los medicamentos que dan Gynecomastia?

Haloperidol

Spironolactone

Gynecomastia+ galactorrhea

Verapamil = gynecomastia + bloody nipple discharge.

Grupo 2 of HTN). Peripherally Acting α Adrenergic blockers-

Mneumomic = De zosin para Tete

- Doxazosin
- Prazosin

- Terazosin
- Tamsulosin = Choice for BPH

Todos se usan prophylactic =(para que no te de) - Autonomic Dysreflexia. = pero si ya la tienes el med for Autonomic Dysreflexia= Nifedipine = Choice.

Side Effect

1. Drowsiness
(Orthostatic hypotension)
2. GI Disturbance
3. Water retention opposite to Na retention (weigh gain, edema)

Grupo 3 of HTN). Centrally Acting sympatholytic

- Clonidine
- Guanfacine
- Methyldopa = Choice for pregnant
-

Nota: los unicos med que puedes usar e embarazada hipertensa.

- LOL=(metoprolol)
- Metildopa
- Hydralazine = (only in emergency)
-

Nota: Magnesio sulfate: No es un HTN. = is for seizure, decrease clonus, use for pre-eclampsia-eclampsia, and to stop labor.

BOARD: pregnant le pusieron MGsulfate, pt ya no tiene seizures pero le queda la presion alta que le doy = Hydralazine.

Side Effect

1. Orthostatic hypotension (drowsiness, dizziness)
2. Sodium & water retention = weight gain & edema
3. Impotence.
4. Depression
5. Dry mouth

Grupo 4 HTN. Angiotensin- Converting Enzyme (ACE Inhibitors)

Angiotensin – Converting enzyme	Angiotensin II Receptor Blockers
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ACE Inhibitors (Priles)	ARBs (Sartan)
• Captopril • Enalapril • Lisinopril • Ramipril Renal protective, lo protegen al rinon. but if you have already renal problems, they are contraindicated. S/E only for PRIL -Taste alteration= dysgeusia= decrease el gusto -Dry caught, = 1 month is expective but-if persist for more than 2-3 months call dr to change medication.	• Losartan • Irbesartan • Valsartan • Candesartan

Side Effect de todos

1. ↑ K (no with K sparing)
2. Orthostatic hypotension
3. Angioedema = (es toda la cara inchada)
4. Tachycardia
5. Headache
6. Insomnia
7. Dizziness
8. Fatigue
9. Bleeding (petechia, browsing)
10. Fluid retention

Grupo 5 of HTN. β -Blockers

- Use for HTN
- Cardiac protective

Non-Selective	Cardio selective
Bronchospasm - Nadolol - Pindolol - Sotalol - Timolol (=glaucoma) - Carvedilol - Propranolol (choice=thyroid storm)	Safety in asthma/COPD - Atenolol - Acebutolol - Betaxolol (=glaucoma) - Bisoprolol - Metoprolol Nota: AMBA = ASTHMA

Side effect

1. Bradycardia (check for HR if less than 60 do not administer) less than 60 withhold and call dr. BOARD question

No in bradycardia

No in heart block

Mask hypoglycemic sign = no in diabetic

Da impotency.

2. Hypotension
3. Dan Bronchospasm (NO AMBA)=Asthma= no dan bronchospasm. Son safety in asthma
4. Dizziness, weakness, fatigue
5. Mask hypoglycemia symptoms
 - ☐ No in diabetes
6. Impotence
7. Agranulocytosis (risk for infection)
8. Depression/psychotic respond
9. Nightmare, Psychotic response

Grupo 6 of HTN. Calcium channel Blocker

1. Diltiazem =Choice for (Atrial flutter/Atrial fibrillation)
2. Verapamil =Choice for (Atrial flutter/Atrial fibrillation)
3. Nifedipine = choice for autonomic dysreflexia and as a tocolytic on preterm labor
4. amlodipine
5. nicardipine

Nota: med use for preterm labor: as a tocolytic to stop labor

- Terbutaline = for asthma. S/S = hyperglycemia no in diabetic
- Nifedipine
- Magnesium sulfate for pre-eclampsia eclampsia
- Indomethacin =nsaid. S/E GI bleeding. Primo del ibuprofen. use to acute gout acute, and for ductus arteriosus persistent

Use all for:

- ☐ Choice for: vasospastic or Prinzmetal angina and Raynaud disease= Vasoconstriction
- ☐ Choice for: Cardiac dysrhythmias (Atrial flutter/Atrial fibrillation)
- ☐ Promote vasodilation

Side effect

1. Bradycardia = check heart rate
2. If less than 60 withhold med and call Dr.
Bajan el pulso y la presión.
3. Verapamil: Gynecomastia and Bloody nipple discharge
4. Dizziness, lightheaded

5. Hypotension
6. Headache
7. Peripheral Edema
8. Constipation
9. Flushing of the skin
10. Change in liver and kidney function.

Grupo 7 HTN. Direct Acting Vasodilators

1. Hydralazine: Use only in emergencies under supervision, son los que mas bajan el blood pressure te la ponen practicamente en cero.)
2. Sodium Nitroprussiate. = (use in Emergency) Covered by a dark or Ambar bag (bc photosensitivity) (la luz lo inactivate) Is stable for 24 hours (discard after 24 hours) -Discard the medication if red, green, or blue. bc se inactivo y ya no sirve)

Board: Patient have pre-eclampsia = tto with magnesium sulfate (bc clonus & seizure) but still HTN, what to do? = Hydralazine

Side Effect (both)

1. Orthostatic hypotension (=Administer AT bed)
2. Severe hypotension
3. Fluid retention, edema
4. Dizziness
5. Headache
6. Nasal congestion
7. Neurologic symptoms (confusion, LOC alteration) (bc el brain se quedo sin O2)

Nota: todos los medicamentos para la presión tienen como side effect = Drowsiness. Para el pheocromocytoma el choice es phentolamine, pero si no esta = choose hydralazine.

Grupo 8 HTN. Miscellaneous vasodilators

1. Nesiritine – Use for uncompensated CHF
CHF = Diagnosis = $BNP > 100$ = Type B natriuretic peptide.

Nota: for CHF choice = digoxin when compensate

Side effect

1. Hypotension
2. Confusion
3. Dizziness
4. Cardiac Arrhythmias

Grupo 5 Cardio. Positive Inotropic & Cardiotonic medication

- Dopamine
- Dobutamine
- Milrinone= board

Choice for Shocks

Side effect

Hypotension: rebound hypotension

Thrombocytopenia= risk for bleeding

Dysrhythmias

Nota: milrinone is opposite to midodrine

Milrinone No increase BP BP = ↓	Midodrine In crease Use for increase BP Side effect ☐ Chills. – tingling. ☐ Urge to urinate goose bump (pelos parados)
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G6. Digoxin = Choice for CHF

Mechanism Action

Increase contractility of myocardium.

☐ Inhibit Na/K pump which causes the heart muscles fibers to contract more efficiently

☐ Therapeutic level

0.5 – 2.0. more than 2 = toxicity. (K bajo aumenta el risk de toxicity)

Sign of toxicity

- Halo vision: yellow, green/ Diplopia
- Nausea, vomiting, and diarrhea
- Blurred vision, photophobia
- Bradycardias (less than 60 in adults or 90-100 in children's = stop and call dr)
- *Contraindicate in
 1. AV block
 2. Bradycardia
- Drowsiness
- Fatiga
- Weakness

Increased Risk Digoxin toxicity (todo lo que baje el potassium)

☑ $K \downarrow$ = All diuretics except K sparing

☑ Sodium polystyrene (kayexalate)

☑ Steroids

☑ Cushing

☑ Licorice

☑ Ginseng

Nota: suben el potassium?

= Captopril

= Enalapril

= Sartan

Grupo 7 de Cardio. Antidysrhythmic Medication

1. Sinus Bradycardia (HR<60) Never give Digoxin – CCB or LOL <u>Treatment</u> Subirle el hearth rate Atropine (if symptoms) S/S Syncope/Dizziness - Pacemaker (si no mejora con tto) Nota: bradycardia del atleta no lleva tto	2. Sinus Tachycardia (HR= 100-150) <u>Treatment</u> Treat the cause (eg dehydrated = IV fluid) - Vasovagal maneuvers (to $\downarrow$ HR) <u>Digoxin</u> (antidote = Digoxin Immune Fab) <u>LOL</u> = choice <u>Diltiazem & verapamil</u> <u>Nifedipine</u>
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Atropina = Anticholinergic (cierra chorro) Sinus Bradycardia 	Sinus tachycardia 
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3. Supraventricular Tachycardia (HR>150) <u>Treatment</u> 1. Vasovagal Maneuver. (ocular pressure, carotid massage unilateral, Valsalva maneuver= bearing down) 2. Adenosine (choice) Administer in IV push In 2 second + 10 ml NaCl 0.9% At bed side Epinephrine, (bc paro cardiaco) Nota: adenosina da como S/E = Asystolia Si no mejora: 3. Diltiazem/Amiodarone 4. Cardioversion (with informed consent) 5. Catheter Ablation	4. Atrial Flutter/ Atrial Fibrillation 1. Diltiazem/= Choice 2. LOL 3. Digoxin, contraindicate in bradycardia, V.tach and AV Block 4. Amiodarone, no in A.VBlock 5. Ibutilide= board, class 3 Antidysrhythmic med,= choice for Atrial Flutter and Atrial fib. S/E Bradycardia, QT prolonged. 6. Quinidine, =use for V tach 7. Warfarine/heparine = to prevent complications: Clots Pulmonary Embolism Stroke 8. Cardioversion Elective (consent) Before cardioversion = withhold digoxin 48-72h before Before cardioversion do TEE = (to check for clot) Transesophageal Echocardiogram (TEE) If clot = do not do it Before TEE: anticoagulants 1 week before After TEE check for gag reflex						
5. Premature Ventricular Contraction PVC ☑ 1-2 PVC -Tto =Avoid stimulants: • Cocaine • Caffeine • Amphetamin e Board: el EKG es un PVC – respuesta = Quitar estimulantes ☑ PVCs = That's not improve =tto: Amiodarone Lidocaine Procainamide Priority More than 3 in a row (PVCs) Multifocal PVCs More than 6 PVCs in 1h 1-2 PVC after infarction, bc thrombolytic therapy, (romper el coagulo) y lo que indica que es effective o que ocurrio revascularization = PVC, pero llamas al doctor bc after del PVC viene el V-Tach	6.Ventricular tachycardia= Check for pulse=#1 action <table border="1"> <thead> <tr> <th><u>Pulse + Stable:</u></th> <th><u>Pulseless</u></th> </tr> </thead> <tbody> <tr> <td> 1. Amiodarone 2. Lidocaine 3. Procainamide 4. Quinidine </td> <td> ☑ Defibrillation + CPR (if both choice) ☑ Put Synchronous button OFF </td> </tr> <tr> <td> <u>Unstable</u> ☑ Cardioversion + CPR Emergency. Synchronized button ON </td> <td> Nota Cardioversion= synchronized button ON. = synchronic Defibrillation= Synchronized button off = asynchrony </td> </tr> </tbody> </table>	<u>Pulse + Stable:</u>	<u>Pulseless</u>	1. Amiodarone 2. Lidocaine 3. Procainamide 4. Quinidine	☑ Defibrillation + CPR (if both choice) ☑ Put Synchronous button OFF	<u>Unstable</u> ☑ Cardioversion + CPR Emergency. Synchronized button ON	Nota Cardioversion= synchronized button ON. = synchronic Defibrillation= Synchronized button off = asynchrony
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	7. Ventricular Fibrillation= (arrhythmia cardiac mas grave. No cardiac output TX= Defibrillator + CPR
--	--

Amiodarone Side Effects

B= Bradycardia = Never in brady and A.VBlock

Blue Man: blue gray skin discoloration. =if pt take amiodarone y se pone gris o azul=stop med and call Dr. But teaching that this discoloration can lasting until 6 months or more.

I= Interstitial lung disease. Board: take amiodarone and have Pulmonary congestion, SOB, stop and call Dr bc esta damage the lung.

T= Thyroid problems: hypothyroidism and hyperthyroidism, any problem with T3 and T4 stop and call Dr.

C= Corneal microdeposits= halo vision or photophobia stop and call Dr

H= Hypotension:

Hepatotoxicity

QT prolonged = Priority bc after QT prolonged viene Arrhythmia ventricular= grave

El que mas photosensitivity da

Lidocaine

No in liver problems (bc liver toxicity)

No in renal problem

☐ If Toxicity:

- ☐ Respiratory Depression
- ☐ LOC alteration, confusion
- ☐ Seizure

Board: Que chequear before or preguntar before lidocaine? = Assess for dentist anesthetic allergies.

Procainamide (ojo)

☐ No in LES(lupus)

☐ No in Myasthenia Gravis

Side Effect

1. Hypotension
2. Nausea, vomiting, diarrhea
3. Dizziness

Quinidine

- PO with glass of water
- ☒ No with grapefruit

Side Effect

Cause Passing out and fainting

- Diarrhea
- Hearing loss/Tinnitus
- Flushing
- Depressed mood
- Vision change
- Peeling skin
- QT prolonged = Torsade de pointe

G3 = Psychiatric

Grupo 1. Mood stabilizer

Lithium = Choice for bipolar for phase (maniac)

Therapeutic level = 0.5 – 1.5 meq/l (+than 1.5=toxicity.)

Side effect

☒ Expected Do not stop medication:

1. Mild or fine hand tremors for 5-7 days after treatment = Expected if persist more than 5 days or stronger = yes Call the Dr.=toxicity
2. Increased urinary output
3. Thirst= (darle agua)

☒ Signs of Lithium Toxicity (OJO) = more than 1.5

Early Toxicity	Late Toxicity= neurologic
1. Coarse hand tremors 2. Mild or fine hand tremors persistent for more than 1 week 3. N/V, diarrhea. 4. Muscles weakness 5. Tinitus 6. Blurred vision	1. Persist GI upset more than 1 week.N/V/D 2. LOC alteration, confusion, ataxia, slurred speech, seizure, coma 3. Visual and tactile hallucinations 4. Oliguria or Anuria.

Board: If do not specific, patient with lithium and complains of hand tremors = Stop + Call the Dr.

Nursing Intervention (lithium ↓ Na levels) (goal =maintain Na levels between 135-145)(bc mientras mas bajo sea el Na mas toxico es el lithium).

1. Take with food (bc GI upset) + plenty fluids
2. Labs to check =Board SATA. Check all less than Liver

Nota. Med que respetan el liver: lithium and aminoglycoside= gentamicin, kanamycin.

1. Na Level (normal = 135 – 145) (mantener Constante. =Allow table salt.)
2. Thyroid function
3. Renal function (BUN + Creatinine)
4. Lithium levels = Check every 2 weekly
 - a. After 12 hours of last dose

5. Cardiac Function - Most serious side effect = Cardiac Arrhythmias

3. Allowed Table salt = OK. (no se aumenta pero no se elimina)

- a. Do not use substitute salt.

Nota: Addison hay que increase salt intake.

4. No with low Na Medication
 - a. No with Diuretics
 - b. No with aminophylline
 - c. No with sodium bicarbonate
 - d. No with coca- cola
5. During treatment avoid Hot temperatures (bc sudor)
6. No in pregnant (bc malformation congenic)
7. No in breast feeding.
8. Don't run

Grupo 2. Anxiolytic Medication (Lisis de la ansiedad)(anxiolytic opposite to SSRI)

To treat anxiety disorder

- Phobias
- Post traumatic stress disorder
- Conversion disorders

Board: con cual de ellos el Pt puede manejar: Buspirone Buspar

G1. Benzodiazepines (Pams) calman la ansiedad.

1. **Diazepam** = Choice for status epilepticus. That is the emergency in seizure.
If intolerance call Dr to change for Buspirone
2. **Lorazepam**: se puede dar para el withdraw del alcohol.
3. **Clonazepam**
4. **Oxazepam**: Choice for liver cirrhosis, bc no se metaboliza en el liver
5. **Chlordiazepoxide** (choice for alcohol withdrawal)

NOTE: Medazolan: Intrahospitalario para relajarte en la cirugía

Benzodiazepines side effect

Take for short time, /no in elderly

1. Addiction
2. CNS Depression
3. Drowsiness/ somnolence
4. Orthostatic hypotension
5. Tolerance
6. Blood Dyscrasias ($\downarrow$ WBC, $\downarrow$ RBC, $\downarrow$ Platelets)
7. Ataxia
8. Confusion
9. Anticholinergics = No in glaucoma/ No in BPH

Overdose. Of benzo (Life-threatening)

1. Respiratory Depression
2. CNS Severe depression

Antidote (Benzodiazepines)

Flumazenil

No with antianxiety herb

1. Chamomile
2. Kava
3. Melatonin
4. No con ballerina
5. No with alcohol
6. Take at bedtime

El overdose de las tablets lleva: gastric lavage with activated charcoal

En los líquidos: give meal or water

Nota: Pt with pain renal calculi le pusieron morphine y se fue manejando, tuvo un accidente y familia demandó al hospital, que se violó? = Non-maleficence principle

G2. Non Benzo

1. Buspirone

- No addiction/ No tolerance
- Side effect: Dizziness or lightheaded
- Little sedation
- Patient can drive

2. Zolpidem

- Choice for insomnia/ Take at bedtime (on bed) porque te duerme

3. Ramelteon:

- Use for insomnia

Grupo 3. Antidepressant Medication

1. Tricyclic Antidepressant
2. MAOIS
3. SSRI
4. SNRI
5. NDRI
6. Atypical Antidepressant
7. St John wort
8. Electro convulsant therapy.

G1. Tricyclic Antidepressants (son los unicos que se toman en la noche)(at bedtime)

1. Amitriptyline/ Nortriptyline
2. Clomipramine = choice for OCD
3. Desipramine = mal uso = bajar de peso
4. Imipramine = choice for nocturnal enuresis (ninos que se orinan en la cama) and urinary incontinence
5. Doxepin

Side Effect

1. Cardiac arrhythmias – EKG monitoring before and during.
 - QT prolonged
 - Torsade de pointe
 - Atrial fibrillation
2. Orthostatic hypotension = take at bedtime
3. Anticholinergic side effect = no un BPH -no in glaucoma
4. Gynecomastia (same to spironolactone, haloperidol and verapamil) + galactorrhea expect verapamil that cause bloody nipple discharge)
5. Sexual disfunction = Decreased libido, decreased erection.
6. Photosensitivity

Board: Nurse de ICU for psychiatric what Pt give to her: Alcohol withdraw or overdose of medication

Overdose = Life threatening

Intervention:

1. Gastric lavage with activated charcoal Nota: same to all PO overdose
2. Neostigmine/ Physostigmine = (same to myasthenia gravis)
 - Physostigmine choice for overdose bc is cholinergic to revertir los anticholinergics side effects
3. Antidysrhythmic medication

☐ wait 2 – 3 weeks (time between different antidepressant)

☐ Therapeutic effect = 3 – 4 weeks

Note: Para cualquier antidepresivo hay que esperar 2 – 3 weeks para cambiar

G2. MAOIS

Mnemonic = Sele Iso Phe Tranyl

1. Selegiline
2. Isocarboxazid
3. Phenelzine
4. Tranylcypromine

Side effect

1. Hypertensive crises associate with Avoid Tyramine containing food

Avoid

- Avocado
- Banana
- Figs
- Papaya
- Sherry
- Egg plant
- Aged cheese = except Cottage cheese and cream is allowed.
- Red wine
- Beer
- Soy sauce
- yogurt
- Brewer's yeast
- Raising
- Sauerkraut
- Pickled
- Sausage
- bologna
- Salami
- Pepperoni

Hypertensive crisis assessment

1. Pounding headache or neck soreness.
2. Chest pain
3. Nausea/vomiting
4. Neck stiffness
5. Mydriasis

a. Note: BOARD: patient que le comio la comida a otro patient y empieza con alguno de esos síntomas = Crisis Hypertensive.

Treatment (hypertensive crisis)

1. Phentolamine
2. Hydralazine = Choice for autonomic dysreflexia

Other Side effect of MAOIS

1. Constipation
2. Urinary retention
3. Insomnia = Take in the morning
4. Increased weight (fluid retention)
5. Orthostatic hypotension = Change position slowly
6. Impotence (delay ejaculation)

⚠ Wait 2 weeks between antidepressant medication

Nota: Dos semanas después de tomarlo el patient tiene fuerza y hay mas risk for suicidal.

MAOIS no whit:

- SSRI
- Antidepressant tricycling
- St john worst
- No with stimulants
- No with Linezolid, (bc tiene MAOIS en su composicion.)

G3. Selective Serotonin Reuptake Inhibitors (SSRI)

Mnemonic = Es cita para ser flu flu

1. Escitalopram
2. Citalopram
3. Paroxetine
4. Sertraline
5. Fluoxetine
6. Fluoxetine

More safety and choice for depression bc less side effects

Take in the morning bc dan insomnia.

Side effect

1. Decrease Libido in men with impotence (sexual dysfunction)
2. Increased weight in woman
3. Anticholinergic side effect = Dry mouth
4. CNS stimulation/ restlessness agitation
5. Blood pressure change
6. Photosensitivity
7. Insomnia
8. Painful menstruation
9. Headache

Complication or Adverse reaction

Serotonin Syndrome = (Life threatening Emergency) (when you take too much SSRI, or you mix with others ex: MAOIS, tricycling, st john wort)

1. Delirium
2. Agitation
3. Tachycardia
4. Increased sweating
5. Increased temperature
6. Muscle rigidity
7. Myoclonus – Seizure
8. Shivering
9. Mydriasis
10. Coarse tremors

Nota:

- Serotonin syndrome. = SSRI = (treat with Cyproheptadine)
- Neuroleptic malignant syndrome: Anti-psychotic = (treat with baclofen and dantrolene)
- Malignant hyperthermia: =Anesthetic reaction = (treat with baclofen and dantrolene)

Las tres se parecen bastante! Debes buscar el Keyword que esta en el History que te den, que medicamento esta tomando? o si se esta operando.?

Baclofen y dantrolene for muscle relaxante but for SSRI is ciprohectadine

Treatment for Serotonin Syndrome

1. Cyproheptadine (choice)
2. Benzo

3. Baclofen or Dantrolene for muscle rigidity, (but is not the choice)

Resumen

- ☐ Tricyclic Medication Side effect
 - ☐ Anticholinergic
 - ☐ Orthostatic hypotension
 - ☐ Arrhythmias
- MAOIs Side Effect
 - ☐ Hypertensive Crisis
- SSRI side effect
 - ☐ Increased weight in women
 - ☐ Sexual dysfunction
 - ☐ Serotonin Syndrome (complication)

G4. Serotonin Norepinephrine Reuptake Inhibitor SNRI

1. Duloxetine = Use for polyneuropathic pain
 2. Venlafaxine = Use smoking cessation –
- No in seizure disorder bc convulsions

S

side effect

Nausea
 Dry mouth
 Insomnia
 Headache
 Fatigue
 Decrease appetite
 Increase sweating
 Sexual dysfunction
 NO mix with others.

G5. Non-Epinephrine and Dopamine Reuptake Inhibitor (NDRIs)

3. Bupropion = used for smoking sensation.
4. Mirtazapine

Side effect

Insomnia
 Agitation
 Headache
 Dry mouth

G6. Atypical Antidepressant**1. Trazadone****Side effect**

- Priapism (erection pene por mas de 4 horas)
- Blurred vision
- Dizziness
- Drowsiness
- -Headache

Grupo 4. Antipsychotic med**Use for Schizophrenia.****Typical****Typical**

*More common use for **positive sign** of schizophrenia

Sign positive are:

The classic symptoms of the disease:

Hallucination

Dilution

Illusion

Disorganized speech

Mnemonic: **El Loco que va pal zine**

Haloperidol

Fluphenazine

Perphenazine

Chlorpromazine

Trifluoperazine

Except

Phenelzine that is SSRI

More Extrapyramidal side effect

Atypical

*Use more for **negative symptoms** of schizophrenia

Negatives sign are

Same to depression

Anhedonia: = (loss of joy)

Abulia

Apathia

A-socialization

Mnemonic: **Condone para el pine de ari**

Clozapine = Decrease WBC= Risk for Infection

Olanzapine = ↑ cholesterol, triglycerides and glucose

Quetiapine = ↑ cholesterol, triglycerides and glucose

Risperidone = ↑ cholesterol, triglycerides and glucose

Ziprasidone = More prolong QT **but si en diabetes**

Aripiprazole = use in Tourette syndrome = tics

Less extrapyramidal side effect

Clozapine = Decrease WBC= Risk for infection, use for myocarditis

Check CBC panel, blood count (if dolor de garganta stop med and call dr)(if fever stop and call dr)

Board: Pt taking fluphenazine and dr change to risperidone que side effect expect ver menor: Shuffling gate.

Side effect of ALL

1. Anticholinergic side effect

2. Extrapyramidal Side effect = Expected Do not call the Dr.

Same to Parkinson induced (by medication)

- Tremor
- Shuffling gait
- Rigidity
- Drooling
- Dysphagia

☐ Medication for Extrapyramidal Side effect = Anticholinergics

- Benadryl
- Benztropine
- Thyhexiphenidine

Board: Pt with schizophrenia Tx with Haloperidol
+ benztropine for extrapyramidal side effect. Que
efecto debe tener doble? Anticholinergic S/E

3. Dystonia = Call the Dr.

- Abnormal involuntary eye movement = oculogyric crisis
- Twisting of the torso or other muscle group

4. Akathisia = Call the Dr. (jiribilla)

- Restlessness
- Constant movement

5. Tardive dyskinesia – months to years after treatment = irreversible = stop and- Call the Dr.

- Protrusion of the tongue
- Chewing motion
- Involuntary movement of the body and extremities.

6. Other side effect

1. Orthostatic hypotension
2. Drowsiness
3. Blood dyscrasias
4. Decrease WBC, decrease platelets
5. Pruritus, photosensitivity
6. Weight gain
7. ↑ blood glucose (except Ziprasidone)
8. Impaired body temperature regulation
9. Gynecomastia

10. QT Prolonged + torsade de pointe

Complication

Neuroleptic malignant Syndrome Life threatening emergency (ICU admission)

- Is a potentially fatal syndrome that may occurs at any time during therapy (more frequently at the beginning)
- ICU admission

Assessment

1. SOB, tachypnea
2. Tachycardia
3. Fever (bien alta)
4. Increased sweating
5. BP labile (up and down)
6. Skeletal muscle rigidity
 - Rhabdomyolysis =(se rompio el musculo)
 - Muscle pain o Hematuria
 - ↑ CK-MM
7. Loss of bladder control
8. Excessive weakness, fatigue.
9. LOC alteration
10. Severe extrapyramidal S/E
11. Seizure
12. Excessive salivation
13. Oculogyric crisis (ojos virados)
14. Dyskinesia
15. ↑ WBC, ↑ Liver enzymes

Intervention

1. Stop medication immediately and call 911 or Dr.
2. Dantrolene, bromocriptine, baclofen

Nota: Cause of malignant Hypothermia

☐ Anesthetic Reaction

☐ Hereditary

☐ Treatment = ICE

Medication to treat Attention- Deficient- Hyperactivity- Disorder ADHD

1. Amphetamine
2. Atomoxetine
3. Methylphenidate

Take In the morning after breakfast to prevent insomnia

Side Effect

1. Tachycardia
2. Increase BP
3. Anorexia and weight loss – take in the morning after breakfast.
4. Agitation
5. Dizziness
6. Insomnia
7. Dry mouth/ headache

Sedation holiday to prevent tolerance and addiction. (Weekend and summer)

Nota: Sedation holiday para la ventilator associate pneumonia, every day at the same time te quitan los sedativos para ver tu respuesta y ver si te pueden quitar la maquina.

Medication to treat Alzheimer disease

Mnemonic: don Roberto Galan que me mantiene

1. Donepezil

Side effect

Mild to moderate dementia

- ☐ Nausea/ vomiting/diarrhea
- ☐ Bradycardia with vagotonic effect= (se desmaya) (safety)

2. Galantamine

Side Effect

Mild to moderate dementia

- N/V, diarrhea, weight loss
- ☐ Anorexia
- ☐ Can cause Bronchoconstriction
- ☐ No in Asthma/COPD

2. Memantine

Side effect

Moderate to Severe dementia

- ☐ Dizziness
- ☐ Confusion
- ☐ Headache

- ? GI disturbances N/V/D (side effect common en todos)

3. Rivastigmine

Moderate to severe dementia

Side effect

- ? GI disturbances
- ? No in peptic ulcer disease
- ? No in bradycardia
- ? No in COPD/ Asthma

Maternity and Newborn Medication

G1. Tocolytic:

Use for preterm labor (to stop)

? Indomethacin

? NSAID. Use for

1. Tocolytic = to stop preterm labor.
2. Ductus Arteriosus persistent
3. Acute gout

Side Effect

- GI bleeding: (no in gastritis no in peptic ulcer)
- Maternal = N/V, dyspepsia=(digestion lenta)

3. Nifedipine for

For blood Pressure

Side effect

1. Orthostatic hypotension
2. Dizziness
3. Headache
4. Facial Flushing
5. Nausea

4. Terbutaline (SABA) for Asthma

Side Effect

- Tachycardia/Restlessness
- Tremor
- ↑ BP, **no in hypertension or cardiovascular disease**
- ↑ Blood Glucose = **No in Diabetes**
- ↓ K (no with digoxin)

. **Magnesium Sulfate.** (Therapeutic level in pregnant 4-7)

- Choice for pre-eclampsia eclampsia. (To decrease clonus)= to prevent and treat seizure
- Use as a tocolytic to stop labor

Before giving: check respiratory rate bc is a CNS depressant

Valor normal in blood: 2.5-5

Therapeutic level in pregnant 4-7

- More than 7 in pregnancy is toxicity= cause respiratory depression, RR less 12. = Stop Mg, call dr and administer the antidote= (calcium gluconate)

Note: solo se administra el antidote cuando hay respiratory depression.

De lo contrario solo decrease or stop the infusion and call Dr.

- Decrease or absent DTR: normal DTR = 2, if (1 or 0) = decrease or stop med and call doctor
- Decrease urinary output

G2. Induction of labor or Ripen the Cervix

1. Prostaglandins (use when Bishop Score ≤ 4)

(don't use when B-shop score > 6)

1. Prostaglandin E1 = Misoprostol (Use for abortion) (bc es dilatador del cuello.)
2. Prostaglandin E2 = Dinoprostone (cervical vaginal Insert)
3. Prepidil o cervidil gel.

B-shop score is a score of OB associate with dilation, that said if patient is ready to parir o no) se pone cuando no haz dilatado.

Side Effect

1. GI Side effect (diarrhea, nausea, vomiting, stomach cramps)
2. Fever chills, flushing headache, hypotension.
3. Hyperstimulation of the uterus.
4. Fetal passage of meconium

Contraindications

1. Active Cardiac, hepatic, pulmonary or kidney disease.
2. Acute pelvic inflammatory disease
3. History of Cesarean
4. Maternal fever or infection
5. Placenta previa

2.Oxytocin (Pitocin)

1. Stimulate the smooth muscle of the uterus and ↑ uterine contraction.
2. Promote milk let down.
3. Uses for induction
4. Control post-partum bleeding
5. Use for incomplete abortion

Medication is: Use in first stage of labor to stimulate contraction when B-shop <4 and after placenta expulsion = (3rd stage of labor) para escurrirlo, contraerlo y coja tono para que no bleeding.

if placenta previa contraindicate oxytocin.

Side Effect/ Contraindication

1. Uterine Tetany =uterine contraction during > 90 second. Stop oxytocin to prevent rupture.
2. Hypotension
3. Pulmonary congestion (No in pneumonia lungs fluid)
4. Water intoxication (Na↓ = Seizure)

G3. Postpartum hemorrhage**1. Ergot Alkaloids (choice)**

- Ergonovine
- Methylergonovine

2. Prostaglandin Analog

- Carboprost= (Hemabate)

Cause bronchospasm no in asthma o COPD

3.Oxytocin

Side Effect (associate with vasoconstriction)

1. Tachycardia= (No in cardiac disease)
2. Increase BP
3. Cardiac arrhythmias
4. Myocardial Infarction

5. Angina
6. Seizure

G4. Opioid Analgesic (se pregunta si lo quieren)

Call Dr. if RR < 12

* Se pone in **active phase of first stage** when dilation cervical de 4-7.

No in **latent** because retrasa la dilatation y **no** en la **transition** bc sufrimiento fetal.

1. Meperidine

Side Effect

- Constipation
- Respiratory depression
- Decrease BP
- Diaphoresis o Urinary retention

2. Fentanyl

Side effect

- Same to meperidine
- Drowsiness,
- respiratory depression
- Hypotension

3. Butorphanol (board)

Side effect

- Hypotension
- Confusion
- N/V

Contraindicate if:

- Pt have a localized area infection
- If have malformations congenital de la columna (scoliosis)
- If more than 1 ml of clear liquid in aspiration. =call Dr bc estas en el lugar equivocado

G5. Medication use for Lung Maturity

1. **Dexamethasone**: to stimulate fetal lung maturity
2. **Beclomethasone**

Diagnostic

- **Amniocentesis** a la mama.

L/E normal is 2:1 =less than 2:1 =No lung Maturity

Nota: Estos steroid medicamentos se le ponen a la mama, si la **dexamethasone** no fue suficiente para que los pulmones maduren el bebesito va a nacer con una enfermedad que se llama la enfermedad de la membrane hyaline or acute respiratory distress syndrome, a ese bebesito que ya tiene ese problema es al que se le pone el lung surfactant.

Administer lung surfactant to the newborn with acute respiratory distress syndrome through Endotracheal tube

Lung Surfactant

- Use in case of preterm newborn or respiratory distress syndrome
- Use through endotracheal tube
- Avoid aspiration 1-2 hours after medication instillation

Med to the newborn

Eye prophylaxis

*To prevent ophthalmia neonatorum

1. **Erythromycin = (choice)** (chlamydia & Gonorrhea)
2. Tetracycline (chlamydia & gonorrhea)
3. Silver nitrate (gonorrhea)

Note: administer after bounding with mother

Vit K = phytonadione

To prevent bleeding

Se pone IM middle third of **vastus laterals.**

Musculoskeletal Medication

G1. Skeletal Muscle relaxants: (actúan a nivel central) (te relajan demasiado)

Risk for fall, drowsiness, priority is **safety**, don't drive.

1. Baclofen: used in neuroleptic malignant Syndrome/ hyperthermia malignant

Side effect

- ☐ CNS effect decreased
- ☐ Drowsiness (safety)
- ☐ Dizziness
- ☐ Weakness
- ☐ Fatigue
- ☐ Urinary retention
- **No in renal or hepatic dysfunction**
- **No in seizure disorders**

2. Cyclobenzaprine

Side effect

- Drowsiness

No with MAOIs

- ☐ Anticholinergic S/E
- ☐ No in glaucoma
- ☐ No in BPH
- ☐ Only use for short periods 2 – 3 weeks

3. Dantrolene = Choice for Neuroleptic Malignant/ Malignant Hyperthermia

Side effect

- Liver Damage
- GI bleeding
- **Impotence**
- Photosensitivity
- Drowsiness

4. Methocarbamol

- No in renal Problems
- Urine Discoloration o Brown, black or green = Expected (como el rifampin, piridium)
- Drowsiness, Dizziness
- Seizure
- Anaphylactic reaction (HIVES) = Call the Dr.

Gout

Treatment:

Acute Gout: Podagra (dolor e inflamacion en el dedo gordo del pie)

Colchicine

Indomethacin

Allopurinol + probenecid = to decrease uric acid, cause synergistic effect =
potencial el efecto cuando se dan juntas.

Cronic gout= (tophi= cristales de acido urico que se depositan en la oreja en los
cartilagos)

Allopurinol + probenecid

Allopurinol no in pt with sulfa allergies bc is a sulfa derivate

Antibiotic medication

G1. Aminoglycosides Board (no afecta el liver)

1. Amikacin
2. Gentamicin
3. Neomycin = treatment for Ammoniacal Encephalopathy and before Liver Biopsy
4. Streptomycin = Treatment for TB
5. Tobramycin = Treatment for Conjunctivitis

- **Peak level** (cuando el efecto es mas potente) = 30 minutes after medication administration

- **Through level** (cantidad de medicamento que tienes acumulado en tu cuerpo) = 30 minutes before medication administration after 3 – 4 dose medication.

Nota: el through Depende de la route = (IV + rapid – PO + lento)

Side effect

1. Nephrotoxicity
 - ☐ ↑creatinine
 - ☐ ↑ BUN
2. Ototoxicity
 - ☐ Tinnitus – Deftness
 - ☐ Balance problems
3. Confusion, LOC Alteration
4. Palpitation, BP changes

G2. Cephalosporin = Cross sensitivity with Penicilins

1. Cefaclor
2. Cefadroxil
3. Cefdinir
4. Cefepime
5. Cefotaxime
6. Ceftriaxone
7. Cephalexin

Side Effect

1. GI disturbances
2. **Pseudomembranous colitis (C-Diff)**
3. Headache
4. Confusion/abnormal thinking
5. Dizziness
6. Paresthesias
7. Hypersensitivity reaction/anaphylaxis
8. Nephrotoxicity
9. **Superinfection = Candidiasis (Oral thrust) (Same to peniciline)**

Note: Medication to treat C-dif: metronidazole, Tambien se le da Fidaxomicin.

G3. Fluoroquinolones

1. Ciprofloxacin
2. Levofloxacin
3. Ofloxacin
4. Norfloxacin
5. Nalidixic Acid

Choice for Anthrax and UTI

Side Effect

1. **Achilles Tendon rupture** (if heel pain = stop and call Dr.) Pt no puede apoyar el pie hasta que no llegue el doctor bc se te puede joder y es grave.)
2. **Steven Jonhson Syndrome** (skin rash)

Nota: Cause No. 1 de Steven Jonhson = Antiseizure medication (Lamotrigine)

3. GI side effect

4. Headache, dizziness
5. Insomnia
6. Depression
7. Bone Marrow suppression (↓ WBC) decrease platelets and RBC

G4. Macrolides

1. Azithromycin = Choice for chlamydial infection
2. Erythromycin = Choice for penicillin allergies
3. Clarithromycin = E. Pillory

Side effect

1. GI Side Effect.
2. Pseudomembranous colitis
3. Superinfection
4. Candida,
5. Thrush = (candida en la boca)
6. Vaginal candidiasis
7. Hypersensitivity reaction = (Hives, rash) – Anaphylactic reaction
8. Fidaxomicin = use to treat C-Diff

Nota: C. Diff treatment is metronidazole and fidaxomicin

G5. Penicillin's = Cross sensitivity with Cephalosporin

1. Penicillin G Benzathine: (prophylactic de la fiebre rheumatic)
2. Penicillin G Potassium (crystalline)
3. Penicillin G Procaine (rapilenta)
4. Amoxicillin
5. Ampicillin
6. Piperacillin: Board
7. Oxacillin

Side Effect

1. N/V, diarrhea
2. Fever, maculopapular rash= se pone rojo (with Ampicillin or amoxicillin) Expected = No se contraindica (no se suspende)
3. Anaphylactic Reaction = Urticaria, hives, SOB
4. Nephrotoxicity
5. Superinfection
6. Furry tongue = (pelos en la lengua) Stop and call Dr

Board: What is the diet for patient taking Penicillin?

Diet

- Yogurt
- Butter milk all are prescribed to prevent diarrhea
- Probiotic
- All antibiotics take with empty stomach except = Amoxicillin

G6. Sulfonamide Antibiotics

1. Trimethoprim Sulfamethoxazole. TSM= choice for kidneys.
2. Sulfadiazine= use in burns

Cross sensitivity, tienen sulfa en su composicion y no se dan con estos.

- ASA
- Hydrochlorothiazide
- Furosemide
- Allopurinol
- Celecoxib = board = anti-inflammatory
- Acetazolamide/Dorzolamide = Diuretic de los ojos
- Gli – oral hypoglycemic = diabetes type 2, (glibenclamide, glyburide, glipizide.)
- G6PD (Glucose 6 phosphate dehydrogenase) = es una anemia, No metaboliza la sulfa (no en sulfa)

- Nota: Hepatotoxicity all

Side effect

1. Crystalluria = Give with plenty fluid bc renal insufficiency
2. Hepatotoxicity
3. Photosensitivity
4. Nephrotoxicity
5. Bone marrow depressant
6. GI side effect
7. Steven Johnson syndrome (if skin rash = Stop + Call Dr.)

G7. Tetracyclines

1. Tetracyclines
2. Demeclocyclines = Choice for SIADH S/E more important diabetes insipidus
3. Doxycycline

Side effect

1. Teeth discoloration) Never in children less than 8 years to prevent permanent teeth discoloration) (take with straw)
2. Photosensitivity (no sun exposure during treatment – no tanning)
3. Take with empty stomach with full glass of water, never with milk
4. N/V diarrhea
5. Dry mouth
6. Hepatotoxicity

G8. Miscellaneous Antibiotics

1. Vancomycin = Choice for MRSA
= Use for C. Diff

- Administer slowly in 60 minutes to prevent “Red Man Syndrome” if occurs =
☐ decreased infusion rate. (no se para)
☐ Administer thorough stable vein to prevent extravasation

Side Effect

- Ototoxicity
- Nephrotoxicity= Stop and call Dr if any of this sign
- Thrombophlebitis, = Si ya tiene thrombophlebitis Stop IV, put warm compress and use other brazo para seguir el IV.

Note: En infiltration = stop IV, warn compress y aqui si puedo subir la mano y el IV lo continuo en la misma mano

Hematoma: Stop IV, sube la mano y poner hielo.

Etravasation: Stop IV, remove the IV but keep the line bc introduce a syringe to aspirate 5-10ml of blood and put antidote SQ.

2. Linezolid

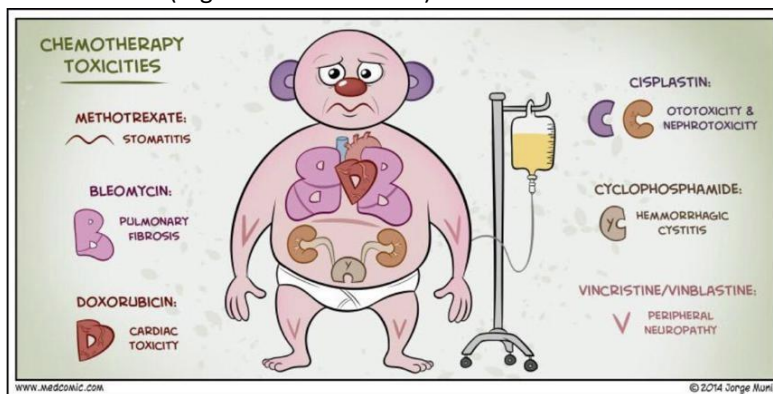
- MAOIs antibiotic, no with MAOIS bc Hypertensive crisis.
- No with tyramine containing food to prevent hypertensive crisis

3. Quinupristin

- Use to treat Vancomycin resistant E.
- VRE/VRS

Anti-Cancer Medication

- Bone marrow suppressant
- Decreased WBC = (Hight risk for infection)



1. Methotrexate (boca)

- Give with Leucovorin to prevent stomatitis.

Side effect

- Stomatitis (bc folic acid deficiency) Treat with Leucovorin
- Liver toxicity
- Complication = Tumor Lysis syndrome (priority) because:
 - $\text{Po}_4 \uparrow$, uric Acid $\uparrow$, K $\uparrow$, Ca $\downarrow$ (todo $\uparrow$ menos Ca $\downarrow$)

Treatment (tumor lysis)

- IV fluid
- Dextrose + Insulin
- Sodium polystyrene = kayexalate for K
- Allopurinol to decrease uric acid

2. Cisplatin (ears and Kidneys)**Side effect**

- Ototoxicity (CN VIII) = Tinnitus
- Nephrotoxicity (kidney) el more Nauseas and vomiting da
- 1. ↑ Creatinine
- 2. ↑ BUN

Tto For vomiting and nausea = Ondansetron o Granisetron (board)

Give 30 min before chemo

S/E ondansetron: Drowsiness, Dizziness, Risk for fall

3. Cyclophosphamide (Blader)**Side effect**

Hemorrhagic Cystitis: give with plenty fluid and *Mezna

4. Bleomycin (Lungs)

Antitumor

Antibiotic o

Side Effect

- Pulmonary toxicity
- Check pulmonary function test before
- If SOB, cough = stop

5. Doxorubicin / Daunorubicin (Hearth)

Side effect

Cardiac toxicity = If any EKG change or cardio problems no se le da.

6. Vincristine / Vinblastine (hands and feet)**Side effect**

- More extravasation
- Peripheral Neuropathy give B6
- Antidote for extravasation= Hyaluronidase

Fuera del Muneco

1. Tamoxifen Citrate

Mechanism of action: = Block estrogen receptors

- Use for breast cancer

Nota: es primo del **raloxifene**

- that is the choice for osteoporosis

Side effects of both

- Thrombosis **DVT** unilateral calf pain
- Endometrial cancer (do ultrasound every 2 -3 weeks

-if bleeding = Stop = call the Dr.

- Liver toxicity

Anticancer Metabolite

1. Fluracil

Side effect

- Alopecia = Expected
- Stomatitis
- Diarrhea
- Photosensitivity
- Tumor Lysis

2. Hydroxyurea

- Preventive of sickle cell Anemia crisis
- Use for Polycythemia vera = (Hight disease)

Side Effects

- Decrease WBC
- **Risk for leukemia**
- No in Pregnant

Monoclonal Antibodies (MABs)

1. **Infliximab** = use for arthritis, Crohn and colitis

2. Adalimumab

☐ Use for Arthritis

☐ Crohn Disease

3. Palivizumabs

☐ Preventive for RSV bronchiolitis (contact Precautions)

4. Omalizumab

☑ Preventive for Asthma

Side Effects

1. Decrease WBC
2. No in active infection
3. No with live virus vaccine

☑ MMR (rubella, rubeola, sarampión, Paperas)

☑ Chicken pox

☑ Nasal influenza

4. No with Immunoglobulins
5. Risk for cancer
6. Anaphylactic reaction

MAB = subcutaneous – 1st Dose under supervision

Colony- Stimulation Factor

1. Filgrastim /Neupogen
 - Use To ↑WBC (se da con anticancer)

El punto donde mas baja los WBC in cancer se llama **Nadir**

2. Epoetin α or Darbepoetin

* Use To increase the Eripoetina y a su vez estimula la produccion de RBC

Use for Chronic Kidney disease = **No in Bone Marrow suppression.**

- o Staring when HB less than 10.5
- o Stop when HB 11 – 12 to prevent Clots: = thrombosis DVT, stroke, MI

Note: Once is open have to use it

- Discard after 24 hours
- Note: HB < 8 = Blood transfusion

3. Oprelvekin

☑ To ↑ platelets

